

Write and interpret ratio notation

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Explain the ratio 5:2 in a sentence. Copy or complete the first step.

2. Complete the equivalent ratio $7:4 = 28: \underline{\hspace{1cm}}$.

3. Simplify 36:48.

4. Miro says: Miro adds the same number to both parts to make an equivalent ratio. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Explain the ratio 5:2 in a sentence. Copy or complete the first step.

Answer For every 5 of the first quantity, there are 2 of the second.

2. Complete the equivalent ratio $7:4 = 28:\underline{\hspace{1cm}}$.

Answer 16.

3. Simplify 36:48.

Answer 3:4.

4. Miro says: Miro adds the same number to both parts to make an equivalent ratio. Is this correct? Explain.

Answer Equivalent ratios come from multiplying or dividing both parts by the same factor.

Write and interpret ratio notation

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Explain the ratio 5:2 in a sentence.

6. Check this result using an inverse, estimate or known fact: Simplify 36:48.

2. Complete the equivalent ratio $7:4 = 28: \underline{\hspace{1cm}}$.

7. Prove or explain your reasoning: Paint is mixed red to white in the ratio 3:5. How much white is needed with 18 ml red?

3. Simplify 36:48.

8. Identify and correct this misconception: Miro adds the same number to both parts to make an equivalent ratio.

4. Solve this independently and show every stage: Explain the ratio 5:2 in a sentence.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Complete the equivalent ratio $7:4 = 28: \underline{\hspace{1cm}}$.

10. Write and solve a similar question to: Simplify 36:48.

Teacher answers and checking prompts

1. Explain the ratio 5:2 in a sentence.
Answer For every 5 of the first quantity, there are 2 of the second.
2. Complete the equivalent ratio $7:4 = 28: \underline{\hspace{1cm}}$.
Answer 16.
3. Simplify 36:48.
Answer 3:4.
4. Solve this independently and show every stage: Explain the ratio 5:2 in a sentence.
Answer For every 5 of the first quantity, there are 2 of the second.
5. Use a second method or representation for: Complete the equivalent ratio $7:4 = 28: \underline{\hspace{1cm}}$.
Answer Expected result: 16. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Simplify 36:48.
Answer Expected result: 3:4. A valid check must be shown.
7. Prove or explain your reasoning: Paint is mixed red to white in the ratio 3:5. How much white is needed with 18 ml red?
Answer 30 ml. The explanation must show why the method works.
8. Identify and correct this misconception: Miro adds the same number to both parts to make an equivalent ratio.
Answer Equivalent ratios come from multiplying or dividing both parts by the same factor.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Simplify 36:48.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Write and interpret ratio notation

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: Paint is mixed red to white in the ratio 3:5. How much white is needed with 18 ml red?

6. Create a new example that tests the same idea as: Simplify 36:48.

2. Prove the answer to this question, rather than only calculating it: Explain the ratio 5:2 in a sentence.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Complete the equivalent ratio $7:4 = 28: \underline{\hspace{1cm}}$.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 3:4.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro adds the same number to both parts to make an equivalent ratio.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Paint is mixed red to white in the ratio 3:5. How much white is needed with 18 ml red?
Answer 30 ml. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: Explain the ratio 5:2 in a sentence.
Answer Expected result: For every 5 of the first quantity, there are 2 of the second. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Complete the equivalent ratio $7:4 = 28:___$.
Answer Expected result: 16. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 3:4.
Answer One valid reconstruction is: Simplify 36:48. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro adds the same number to both parts to make an equivalent ratio.
Answer Equivalent ratios come from multiplying or dividing both parts by the same factor.
6. Create a new example that tests the same idea as: Simplify 36:48.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.